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Task 71 : Use Scatter Plot to check data relationship (consistency).

Objective

The main objective of a Scatter Plot is to analyze the relationship between two numerical variables and check how consistent or related the data is.

```
import matplotlib.pyplot as plt
import pandas as pd
df=pd.read_csv("statistical_dataset.csv")
plt.figure(figsize=(4,4))
plt.scatter(df['Age'],df['Salary'])
plt.title("Age vs Salary ")
plt.xlabel("Age")
plt.ylabel("Salary")
plt.show()
```

Output :



In my code, I performed a **data relationship analysis using a scatter plot**. First, I imported the required libraries—Matplotlib for visualization and Pandas for handling data. Then, I loaded the dataset (statistical_dataset.csv) into a DataFrame. After that, I created a figure of size 4×4 and plotted a scatter plot using **Age** on the X-axis and **Salary** on the Y-axis. I added a title and labeled both axes to make the chart clear and understandable. Finally, I displayed the plot. Through this process, I analyzed how Age and Salary are related, helping me identify patterns, consistency in the data, and any outliers.

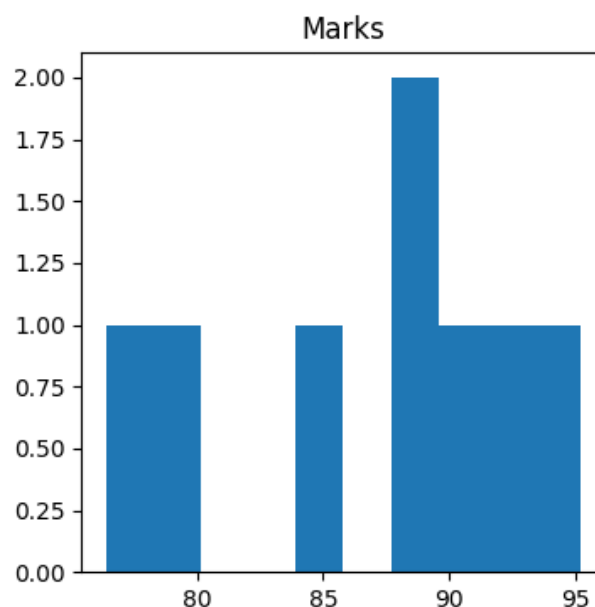
Task 72 : Use Histogram to analyze data distribution.

Objective

The main objective of a Histogram is to understand how data is distributed across different ranges (bins).

```
# 72. Use Histogram to analyze data distribu
df=pd.read_csv("students.csv")
plt.figure(figsize=(4,4))
plt.hist(df['Marks'])
plt.title("Marks")
plt.show()
```

Output :



In my code, I analyzed the **distribution of students' marks using a histogram**. First, I loaded the dataset (students.csv) into a DataFrame using Pandas. Then, I created a figure of size 4×4 for better visualization.

After that, I plotted a histogram of the **Marks** column, which grouped the marks into different intervals (bins) and showed how frequently values fall into each range. I added a title to make the chart clear and finally displayed it. Through this histogram, I was able to understand how the marks are distributed, observe the shape of the data (such as normal or skewed), and identify any patterns or outliers.

Task 73 : Use CSS and JavaScript to create better visuals .

Objective:

The objective of using CSS and JavaScript is to enhance the appearance and interactivity of visualizations. CSS is used to improve styling—such as colors, layouts, fonts, and responsiveness—while JavaScript is used to make visuals dynamic and interactive, like adding animations, tooltips, and real-time updates.

```
<!DOCTYPE html>
<html>
<head>
  <title>Dashboard</title>
  <link rel="stylesheet" href="style.css">
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
</head>
<body>

<h2>Marks Chart</h2>
<canvas id="chart"></canvas>

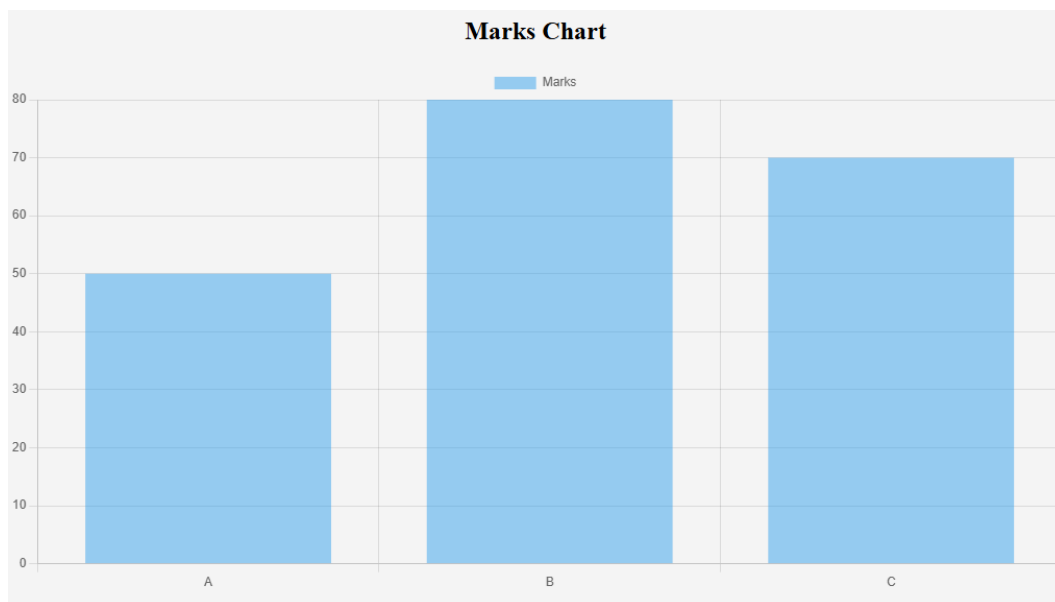
<script src="script.js"></script>
</body>
</html>
```

```
body {
  text-align: center;
  background: #f4f4f4;
}
```

```
const ctx = document.getElementById('chart');

new Chart(ctx, {
  type: 'bar',
  data: {
    labels: ['A', 'B', 'C'],
    datasets: [{
      label: 'Marks',
      data: [50, 80, 70]
    }]
  }
});
```

Output :



In my code, I created a **basic web dashboard structure using HTML** to display a chart. I added a title “Dashboard” and linked an external CSS file (style.css) to style the page.

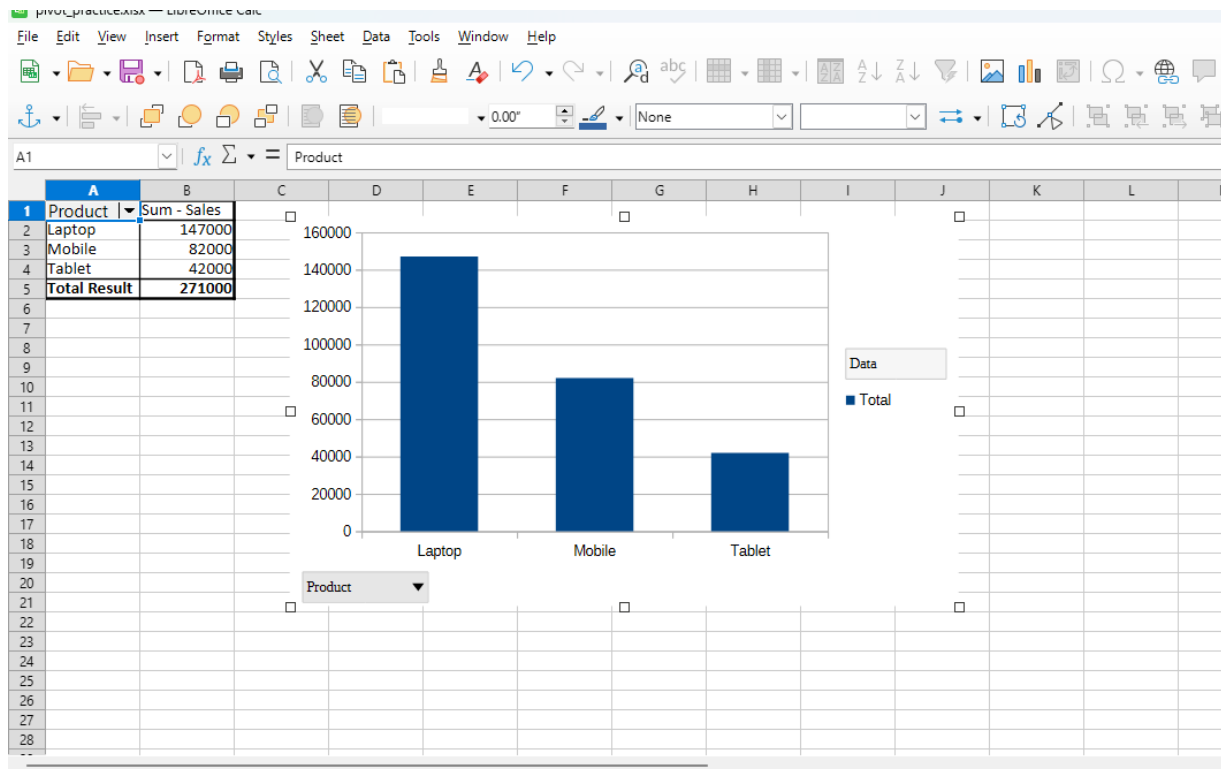
Then, I included the **Chart.js library** using a CDN to enable data visualization. Inside the body, I displayed a heading “Marks Chart” and created a <canvas> element where the chart will be rendered.

Finally, I connected an external JavaScript file (script.js) where the chart logic and data will be written. Through this, I set up the foundation to **create interactive and visually appealing charts using JavaScript and CSS**.

Task 74 : Use Pivot Tables in Excel to visualize data .

Objective

The objective of using **Pivot Tables in Excel** is to **summarize, analyze, and visualize large datasets in a simple and structured way**. Pivot Tables help in organizing data by categories, performing calculations (like sum, average, count), and identifying patterns or trends quickly.



In my work, I used **Pivot Tables in Excel (LibreOffice Calc)** to **summarize and visualize sales data**. I organized the data by **Product (Laptop, Mobile, Tablet)** and calculated the **total sales for each category** using aggregation (Sum).

Then, I created a **bar chart** based on the pivot table, which visually compares the sales of each product. I also included a **total result (271000)** to show overall sales.

Through this, I transformed raw data into a **clear, structured summary and visual representation**, making it easier to compare products, identify trends, and understand which product performs best.

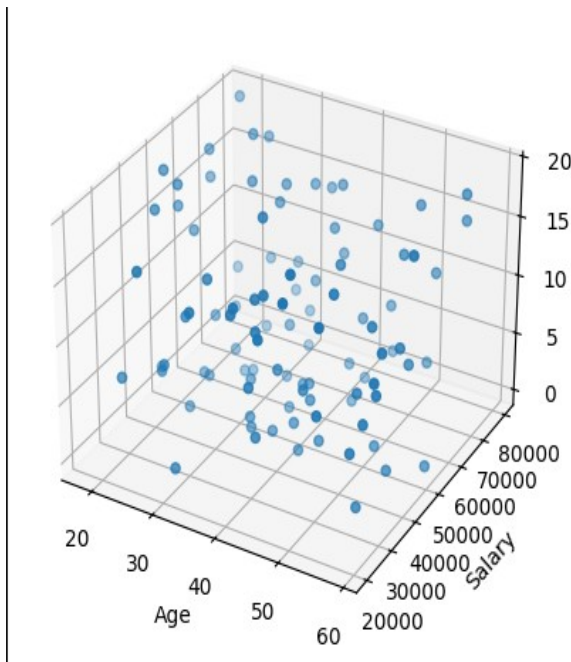
Task 75: Perform 3D Data Analysis (3D Scatter Plot) .

Objective

Visualize the relationship between three numerical variables simultaneously in a three-dimensional space.

```
# 75. Perform 3D Data Analysis (3D Scatter Plot)
from mpl_toolkits.mplot3d import Axes3D
df=pd.read_csv("statistical_dataset.csv")
fig=plt.figure()
ax=fig.add_subplot(111,projection='3d')
ax.scatter(df['Age'],df['Salary'],df['Experience_Years'])
ax.set_xlabel('Age')
ax.set_ylabel('Salary')
ax.set_zlabel('Experience_Years')
plt.show()
```

Output :



I created a **3D scatter plot** to analyze the relationship between three variables from my dataset. First, I loaded the data from a CSV file using pandas. Then, I created a 3D plotting space using Matplotlib and plotted the data points with **Age on the X-axis, Salary on the Y-axis, and Experience Years on the Z-axis**. After that, I labeled each axis for better understanding and displayed the graph.

This helped me understand how salary is influenced by both age and experience at the same time, and made it easier to identify patterns, trends, and outliers in the data.

Task 76 : Determine Which Graphs are Suitable for Data Visualization .

Objective

To understand which graph types are suitable for different kinds of data visualization.

Different graphs are used for different purposes because not all data can be represented effectively using the same graph.

Simple way to remember:

Compare categories → Bar Chart

Show trend over time → Line Graph

Show relationship → Scatter Plot

Show percentage → Pie Chart Show distribution

Show correlation → Heatmap

First, I analyzed the dataset and identified different types of data present in the Titanic dataset. Then I selected appropriate graphs according to the type of analysis required

For example:

- I used a Bar Chart for passenger class because it compares categories.
- I used a Pie Chart for gender distribution because it shows percentage contribution.
- I used a Histogram for age because it shows numerical data distribution.
- I used a Scatter Plot to understand relationships between Age and Fare.

Task 77 : Visualize Data Flow .

Objective

To visualize how data moves from one stage to another in a system.

Raw Data



Data Cleaning



Data Preprocessing



Data Analysis



Visualization



Result

Explanation of each stage:

Raw Data

- Original collected data
- May contain errors, missing values, and duplicate records .

Data Cleaning

- Removes duplicate values
- Handles missing values
- Corrects incorrect data

Data Preprocessing

- Converts data into a usable format
- Performs encoding and feature selection

Data Analysis

- Finds patterns and relationships in data

Visualization

- Creates graphs and charts for understanding results

Final Result

- Provides useful insights and conclusions

Task 78 : Use Data Analysis ToolPak in Excel .

Objective

Perform advanced statistical analysis quickly and accurately without complex formulas.

Quantity		
Mean		6.75
Standard Error		0.75
Mode		5
Median		6.5
First Quartile		5
Third Quartile		8.25
Variance		4.5
Standard Deviation	2.12132034355964	
Kurtosis	-1.24444444444444	
Skewness	0.314269680527355	
Range		6
Minimum		4
Maximum		10
Sum		54
Count		8

I used the Data Analysis ToolPak in Excel to perform Descriptive Statistics on the “Quantity” data. I generated a statistical summary that includes measures like mean, median, mode, standard deviation, variance, range, minimum, maximum, sum, and count.

I also analyzed the distribution of the data using skewness and kurtosis, and identified quartiles to understand how the data is spread. This helped me summarize the dataset, understand its central tendency and variability, and gain insights into the overall distribution of the values.

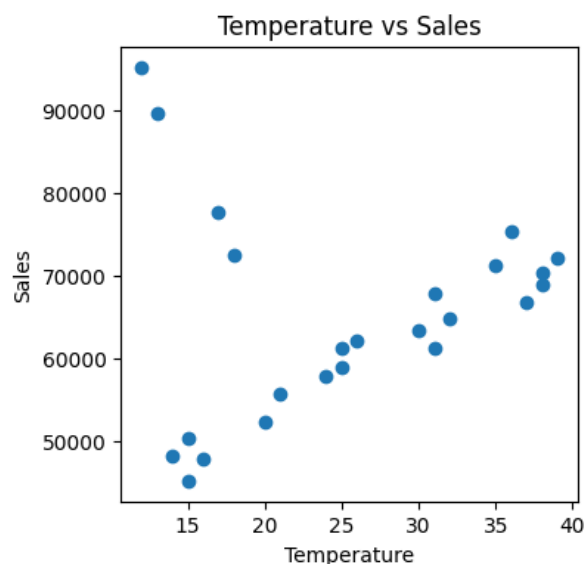
Task 79 : Visualize real-world data points objective .

Objective

Represent actual data in graphical form to better understand patterns, trends, and relationships.

```
# 79. Visualize real-world data points
df=pd.read_csv("sales_analysis_output.csv")
plt.figure(figsize=(4,4))
plt.scatter(df['Temperature'],df['Sales'])
plt.xlabel("Temperature")
plt.ylabel('Sales')
plt.title("Temperature vs Sales")
plt.show()
```

Output :



I created a scatter plot to visualize real-world data from the dataset. First, I loaded the data from a CSV file using pandas. Then, I plotted Temperature on the X-axis and Sales on the Y-axis using Matplotlib. I added labels and a title to make the graph clear and easy to understand, and finally displayed the plot.

This visualization helped me analyze how sales change with temperature, making it easier to identify patterns, trends, and any relationship between these two real-world variables.

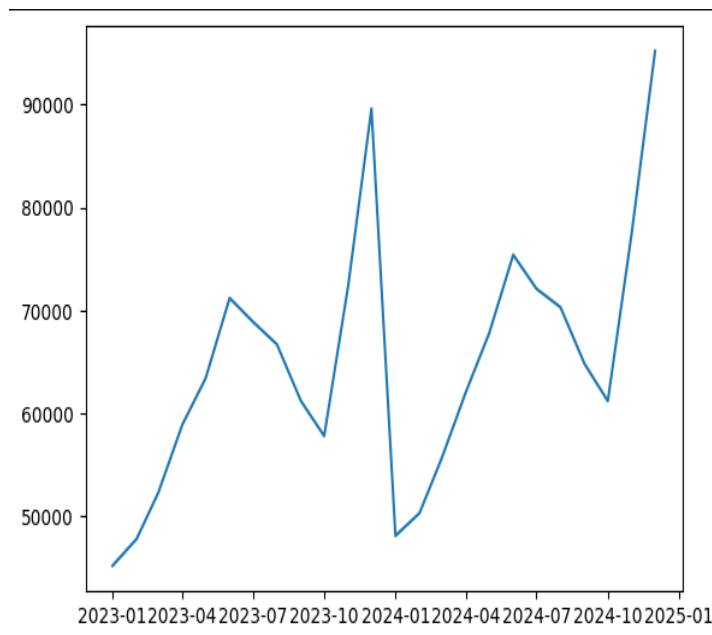
Task 80 : Create Time Series Graph (Time-based data) .

Objective

Visualize how data changes over time to identify trends, patterns, and seasonal variations.

```
# 80. Create Time Series Graph (Time-based data)|
df['Date']=pd.to_datetime(df['Date'],errors='coerce')
plt.plot(df['Date'],df['Sales'])
plt.show()
```

Output :



I created a time series graph to visualize how sales change over time. First, I converted the Date column into datetime format to ensure proper time-based plotting. Then, I plotted Date on the X-axis and Sales on the Y-axis using a line graph. Finally, I displayed the graph.

This helped me analyze trends in sales over time, making it easier to identify patterns, growth, or fluctuations in the data.